

UDAY PRATAP SINGH

FULL-STACK & AI ENGINEER

Email: udaypratap6145@gmail.com | GitHub: github.com/Uday-6145 | LinkedIn: linkedin.com/in/uday-pratap-singh-8a0a40375

PROFESSIONAL SUMMARY

I am a Full-Stack & AI Engineer focused on building scalable, production-grade applications that blend modern web engineering with intelligent systems. My technical foundation centers on the **MERN stack (React, Node.js, Express, MongoDB, React-native)** and **TypeScript**, paired with hands-on expertise in **Generative AI, Retrieval-Augmented Generation (RAG), and Multi-Agent Workflows**.

TECHNICAL SKILLS

- **Frontend:** React.js, React Native, JavaScript, HTML5, CSS3, Tailwind CSS
- **Backend:** Node.js, Express.js
- **Databases:** MongoDB, Mongoose, MySQL
- **AI / Generative AI:** Python, LangChain, LangGraph, RAG, ChromaDB, LangSmith
- **Tools / languages:** Git, GitHub, VS Code, Postman, C++

PROJECTS

Kestrel Multi-Agent Research Assistant | Python, LangGraph, ChromaDB, Groq, LangSmith

GITHUB: https://github.com/Uday-6145/Kestrel_Research_Assistant

- Built a four-agent RAG pipeline: a router interprets the question, a retriever searches ChromaDB, a verifier checks the evidence, and a synthesizer prepares the answer.
- The verifier labels evidence as supported, partially supported, conflicting, or insufficient, and the system avoids guessing when evidence is weak.
- Evaluated 15 questions with LangSmith, reaching 80% verdict accuracy; fixed a citation-parsing bug and moved inference to Groq to improve response speed.

CourseApp | React, Node.js, Express.js, MongoDB

GITHUB: <https://github.com/Uday-6145/Course-selling-backend>

LIVE DEMO: <https://course-selling-backend-puce.vercel.app/>

- Completed a course-selling application with Express and MongoDB backend for managing users, administrators, courses, and purchases.
- Designed the Mongoose data models and organized the REST API into separate routes for the platform's main areas.
- Deployed a live Vercel demo so the application can be reviewed online.

Zyro HR Assistant | Python, LangChain, FAISS, Groq, Streamlit

GITHUB: <https://github.com/Uday-6145/zyro-hr-chatbot>

- Built a question-answering chatbot grounded in 11 HR policy PDFs, so responses are based on the supplied policies rather than general model knowledge.
- Prepared the documents as chunks, embedded them into FAISS, and used MMR retrieval to find relevant passages for each question.
- Added a scope check to screen out questions unrelated to HR policy, plus a Streamlit chat interface with conversation history.
- Included links to source material in the responses so users can check which policy supports an answer.

EDUCATION

NxtWave of Innovation in Advanced Technologies(NIAT) (2025-2029)

B. Tech, SoftwareEngineering | CGPA: 8.75

Kaushalya Devi Public School | Class XII (2024-2025)

Gyan Vallen Co-Ed School | Class X (2022-2023)

ACHIEVEMENTS

- 2nd place, Webathon built a webpage in a three-hour
- Finalist, OpenAI Buildathon.
- Finalist, Smart India Hackathon (SIH) internal round.